

Fitbauer: an open-source Python application for ^{57}Fe Mössbauer spectral analysis with rigorous uncertainty quantification

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Abstract We present **Fitbauer**, an open-source desktop application written in Python for the analysis, simulation, and least-squares fitting of ^{57}Fe Mössbauer transmission spectra. The program is built around *live simulation* of the direct model: every parameter change re-renders the spectrum instantly, so a physically sensible model can be constructed by eye before fitting. It implements discrete fitting of up to six simultaneous components (singlets, doublets, sextets, and magnetic-relaxation lineshapes), as well as hyperfine field distributions $P(B_{\text{hf}})$ and quadrupole-splitting distributions $P(\Delta E_Q)$ with Tikhonov, total-variation, or maximum-entropy regularisation (Hesse–Rübartsch method). Three independent uncertainty-quantification methods are provided: propagated covariance matrices, residual bootstrap Monte Carlo, and profile-likelihood intervals. A rigorous validation suite comprising 34 automated tests at six levels — from mathematical self-consistency to Monte Carlo pull statistics — is included in the distribution. A direct comparison with NORMOS/SITE (Brand 1987) on identical spectra of α -Fe, hematite ($\alpha\text{-Fe}_2\text{O}_3$), siderite (FeCO_3), and magnetite (Fe_3O_4) shows agreement in all hyperfine parameters to within 0.013 T in B_{hf} and 0.004 mm s^{-1} in δ , confirming the correctness of the physical model and calibration. The program is platform-independent (Windows, Linux, macOS), offers an interface in seven languages, and is freely available under an open licence.

Keywords Mössbauer spectroscopy · Spectral fitting · Hyperfine parameters · Distribution analysis · Open source · Python · Uncertainty quantification

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1 Introduction

^{57}Fe Mössbauer spectroscopy remains one of the most direct probes of local electronic structure, magnetic order, and oxidation state in iron-containing compounds. Its widespread use in mineralogy, environmental science, materials science, and geochemistry depends critically on the quality of spectral fitting software used to extract hyperfine parameters from the data [1, 2].

Several programs have been developed over the past four decades. NORMOS [3] and its Windows wrapper WinNormos are the most widely cited; they are written in Fortran and have served the community well but are no longer actively developed and lack modern graphical interfaces. MossWinn [4] (Z. Klencsar, Budapest) is a comprehensive commercial package for Windows only, with outstanding features for thick-absorber correction and a large spectral database. SpectrRelax [5] handles relaxation and distribution models on Windows. Moessfit [6] is a free Qt/C++ program focused on simultaneous multi-dataset fitting and the Maximum Entropy Method. RECOIL [7] (University of Ottawa) and VINDA [8] are further established options, likewise platform-restricted. None of these programs is open source, which limits reproducibility and makes independent code auditing impossible.

Recent years have seen growing demand for Python-based scientific software owing to its ecosystem of well-tested numerical libraries, its suitability for integration into automated pipelines, and the reproducibility of virtual-environment setups [9]. Here we describe Fitbauer, which was designed from the outset with three goals: (i) full separation of the physics engine from any graphical interface, enabling headless (scriptable/CLI) operation; (ii) a comprehensive and formally validated uncertainty-quantification framework; (iii) a modern, cross-platform interface built around *live simulation* of the direct model, available in seven languages.

Section 2 describes the physical model. Section 3 outlines the software architecture. Section 4 summarises the main features. Section 5 presents the validation suite. Section 6 shows application to four reference materials including a direct comparison with NORMOS. Section 7 discusses advantages and current limitations. Section 8 concludes.

2 Physical model

2.1 Lineshape and transmission

Fitbauer evaluates the Mössbauer transmission spectrum in the *thin-absorber* approximation as:

$$T(v) = B + sv - \sum_{k=1}^N A_k(v), \quad (1)$$

where B is the baseline (normalised), s is an optional linear slope accounting for slow background drift, v is the source velocity, and $A_k(v)$ is the absorption contribution of the k -th component.

Each Lorentzian line has the form

$$L(v; v_0, \Gamma) = \frac{(\Gamma/2)^2}{(v - v_0)^2 + (\Gamma/2)^2}, \quad (2)$$

where v_0 is the line centre and Γ is the full width at half maximum (FWHM). A Voigt profile (convolution with a Gaussian of width σ) is also available, evaluated via the Faddeeva function (`scipy.special.wofz`) and normalised to its peak analytically, independently of the velocity sampling.

Spectra recorded with a triangular velocity drive are folded around an automatically refined folding point; sinusoidal-drive data (NORMOS convention `TRIANG=.FALSE.`) are handled unfolded, with each channel evaluated at its true sinusoidal velocity and the drive phase available as a free fit parameter.

2.2 Discrete components

Up to six spectral components can be superimposed; each is one of six types: singlet, doublet, sextet, or one of three magnetic-relaxation lineshapes (Section 2.3).

2.2.1 Sextet and quadrupole treatment

For a sextet the conventional first-order correction applies $\pm \Delta E_Q/2$ (inner/outer pairs). When ΔE_Q is large, an exact Kündig diagonalisation [10] of the excited-state Hamiltonian

$$\hat{H} = \omega_L \hat{I}_z + \frac{\Delta E_Q}{6} (3\hat{I}_z^2 - \hat{I}(\hat{I} + 1)) \quad (3)$$

is available in fixed-angle or powder-average (20-point Gauss-Legendre) variants. Line intensities follow the 3 : 2 : 1 pattern or are parameterised by the texture angle $t = \langle \sin^2 \theta \rangle$.

2.2.2 Thick-absorber correction

For optically thick samples the transmission is modelled as [11]

$$T(v) = b + sv - C(1 - e^{-A_{\text{tot}}(v)/C}), \quad (4)$$

where b and s are the baseline and slope, A_{tot} is the thin-absorber absorption, and C is the saturation scale.

2.3 Magnetic relaxation models

Although the main focus of the program is crystalline (discrete) fitting, three relaxation lineshapes can be combined freely with discrete phases: a phenomenological interpolation between a blocked sextet and a superparamagnetic doublet;

a two-state dynamic model for the $+B_{\text{hf}} \leftrightarrow -B_{\text{hf}}$ exchange following Blume and Tjon [12]; and a Néel–Arrhenius model [13, 14] for polydisperse nanoparticle assemblies (lognormal size distribution, with an optional multi-temperature global fit that shares the physical parameters across spectra).

2.4 Hyperfine field and quadrupole-splitting distributions

For magnetically disordered or nanostructured samples, Fitbauer implements $P(B_{\text{hf}})$ and $P(\Delta E_Q)$ distributions following the Hesse–Rübartsch method [15]: a sum of many sextets (or doublets) on a uniform grid, with amplitudes constrained to be non-negative and regularised by minimising

$$F = \|\mathbf{y} - \mathbf{A}\mathbf{p}\|_w^2 + \alpha \|\mathbf{D}^2\mathbf{p}\|^2, \quad \mathbf{p} \geq 0, \quad (5)$$

where $\|\cdot\|_w$ is the Poisson-weighted norm and $\mathbf{D}^2$ is the second-difference matrix. Besides the Tikhonov penalty of Eq. (5), total-variation (edge-preserving) and maximum-entropy regularisers are available. The regularisation parameter α is selected via the L-curve criterion [16] — with suggestions from maximum curvature, a compromise value, and Generalised Cross-Validation (GCV) — and can be chosen interactively by clicking directly on the L-curve plot. A 1σ uncertainty band on P is computed from the linearised covariance of the weights, and $\tilde{\chi}^2$ uses effective degrees of freedom (trace of the hat matrix). Simultaneous *sharp* (discrete) components can be superimposed on the distribution to handle multiphase samples.

As an alternative to the free-form histogram, parametric shapes can be fitted without any regularisation: a single Gaussian, a binomial (alloy) distribution, and the Voigt-based sum-of-Gaussians of Rancourt and Ping [17], in which the fitted amplitude of each Gaussian directly gives the *area* (population) of the corresponding subpopulation. Two-dimensional distributions such as $P(B_{\text{hf}}, \Delta E_Q)$ are also available, with independent regularisation parameters in each direction.

2.5 Calibration convention

The velocity calibration follows the standard α -Fe convention: $B_{\text{hf}} \equiv 33.0 \text{ T}$ with line positions from published experimental measurements [18] ($\pm 0.839/\pm 3.084/\pm 5.329 \text{ mm s}^{-1}$), not from nuclear moments. Using theoretical positions introduces a systematic error of $\approx 0.1 \text{ T}$ in all fitted fields [19]. Once determined from an α -Fe foil, the calibration is *fixed* and reused automatically for every subsequent raw-counts spectrum, so the velocity scale is defined once per instrument configuration.

3 Software architecture

Fitbauer follows a strict Model–View–Controller separation. The physics and optimisation code lives entirely in the `core/` package, with no dependency on

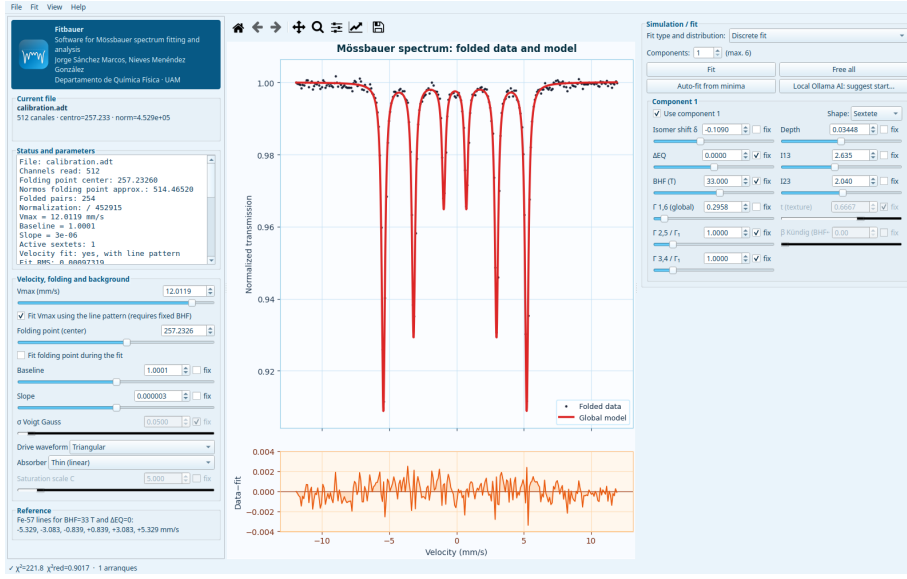


Fig. 1 Fitbauer main window showing a fitted α -Fe calibration spectrum (folded, $^{57}\text{Co}/\text{Rh}$ source). Left: file status, calibration, folding, and background controls. Centre: Matplotlib plot area with data, model, and residuals. Right: simulation/fit panel with the physical parameters of the active component (here a sextet with B_{hf} fixed at 33 T while the velocity scale is fitted).

any graphical toolkit. The Qt front-end (PySide6) and the command-line interface are thin clients of `core/`. This design allows the same computation to be driven interactively, from a Python script, or from a batch pipeline without modification.

- `core/physics.py` Pure functions for lineshape evaluation (Lorentzian, Voigt, thick-absorber), component absorption, and the Kündig Hamiltonian.
- `core/fit_engine.py` Least-squares optimisation via `scipy.optimize.least_squares` (bounded trust-region-reflective), multi-start initialisation, parameter bounds, covariance-matrix uncertainties, bootstrap resampling, and profile-likelihood intervals.
- `core/folding.py` Spectrum loading (WS5 XML and single-column ASCII ADT formats), automatic folding-point search, triangular- and sinusoidal-drive velocity axes.
- `core/session.py` Headless orchestration layer (`HeadlessSession`, `ModelState`): load $\rightarrow$ fold $\rightarrow$ fit $\rightarrow$ export without any GUI.
- `mossbauer_distribution.py` Distribution fitting $P(B_{\text{hf}})/P(\Delta E_Q)$: Tikhonov regularisation, L-curve, and sharp components.

Figs. 1 and 2 show the main graphical interface and a discrete-fit result.

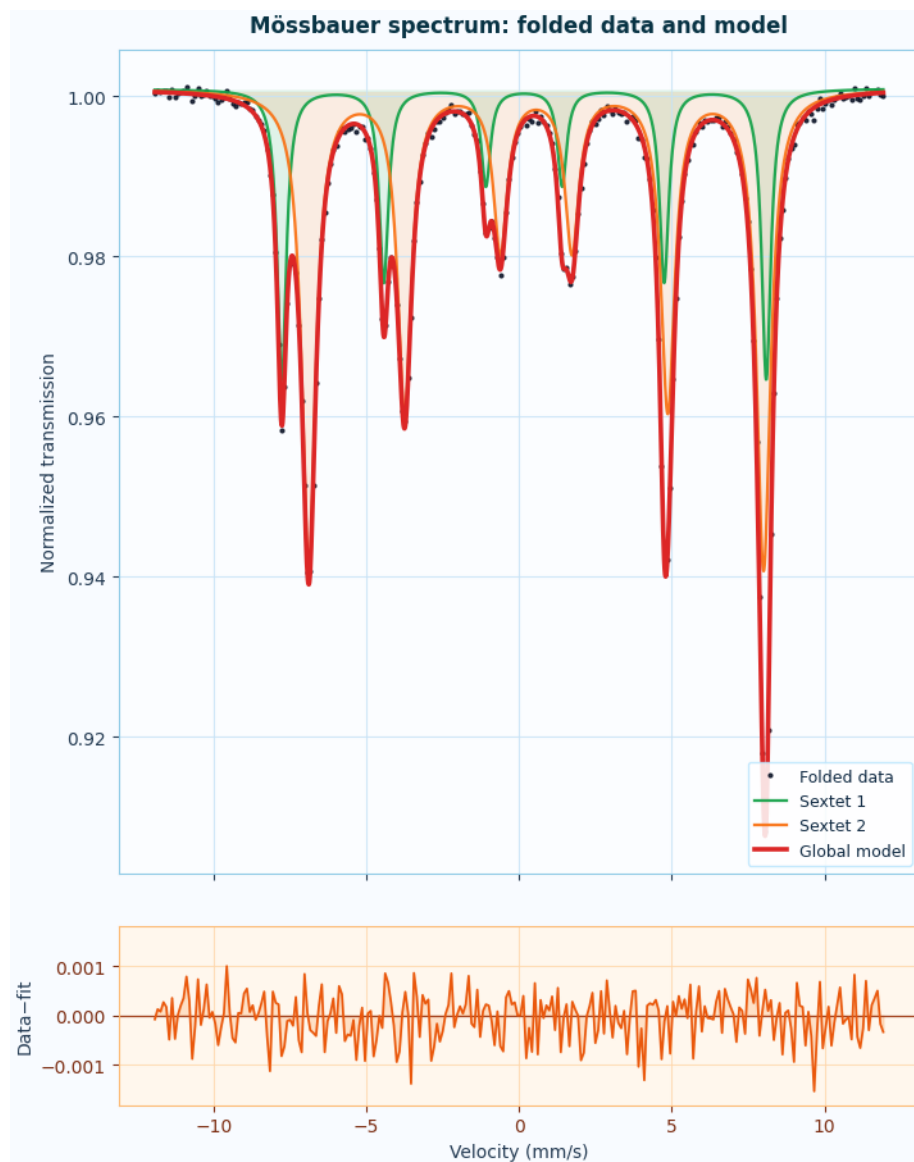


Fig. 2 Discrete fitting of a two-component spectrum (magnetite Fe_3O_4). The two sextet subspectra are shown individually in colour with optional area shading; the total model is the thick red curve; residuals in the lower panel.

4 Main features

4.1 Live simulation of the direct model

A distinguishing design choice is that the direct model is always *live*: every slider or numeric change to any physical parameter (δ , ΔE_Q , B_{hf} , widths, depths, intensities) re-renders the full spectrum and its individual subspectra instantly. This serves several practical purposes: (i) a physically sensible starting model can be constructed by eye before launching the optimiser, which makes the subsequent non-linear fit faster and far less likely to fall into a spurious minimum; (ii) the effect of each hyperfine parameter on the lineshape can be explored interactively, which builds physical intuition and makes the program useful for teaching; (iii) hypotheses can be compared visually — e.g. simulating a two-phase mixture with assumed proportions over a measured spectrum — without fitting anything. Simulation and fitting share exactly the same direct model, so what is simulated is precisely what the optimiser will refine; per-parameter *fix* locks let the user freeze any subset of parameters during this exploration and during the fit.

4.2 Fitting modes

Discrete mode supports up to six simultaneous components of any type, with independent free/fixed flags for each parameter, linear constraints between parameters, and multi-start optimisation (eight Gaussian restarts around the user’s starting point with early stopping, plus an optional differential-evolution pre-pass for multi-sextet label-permutation minima). The noise model can be Gaussian or Poissonian (iteratively reweighted least squares with model-based variances).

Distribution mode (Fig. 3) computes $P(B_{\text{hf}})$ or $P(\Delta E_Q)$ over a user-defined grid, optionally with simultaneously fitted sharp components. The regularisation parameter is selected interactively via the L-curve (Fig. 4) or set manually.

4.3 Uncertainty quantification

Three methods are provided and can be run consecutively on the same fit:

1. *Covariance matrix* (default): symmetric 1σ errors from the Jacobian at the minimum, $\sigma_i = \sqrt{(J^\top J)_{ii}^{-1}}$.
2. *Bootstrap Monte Carlo*: N replicas are generated by adding Poissonian noise at the per-channel level; each replica is independently refitted and the distribution of the N best-fit values gives asymmetric percentile intervals.
3. *Profile likelihood*: the parameter of interest is stepped away from its best-fit value while all others are re-optimised; the 1σ interval corresponds to a unit increase in $\tilde{\chi}^2$ [20].



Fig. 3 Distribution fitting of a magnetically disordered sample (synthetic spectrum generated from a bimodal $P(B_{\text{hf}})$ at 28 and 48 T). Left: fitted spectrum and residuals. Right: reconstructed $P(B_{\text{hf}})$, which recovers the two field populations.

4.4 Other features

Further tools include batch fitting of spectrum series with warm-start from the previous result, a semi-manual absorption-minima editor that proposes starting models from the detected line positions, PDF/Markdown report export, and reproducible session files that store the complete analysis state (data, calibration, model, and results). The interface is available in seven languages (Spanish, English, French, German, Japanese, Russian, and Chinese).

5 Validation suite

The distribution includes `tests/test_synthetic_validation.py`, a 34-test module structured in six levels of increasing rigour (part of a wider automated suite covering the physics core, folding, fit engine, distributions, and the graphical interface).

5.1 Level 1 — Self-consistency

Noise-free synthetic spectra are generated with a reference implementation that is *mathematically independent* of `core/physics.py` (different variable names, no shared code), and then fitted with Fitbauer. The expected result is $\tilde{\chi}^2 < 10^{-6}$ and recovered parameters identical to the ground truth to machine precision.

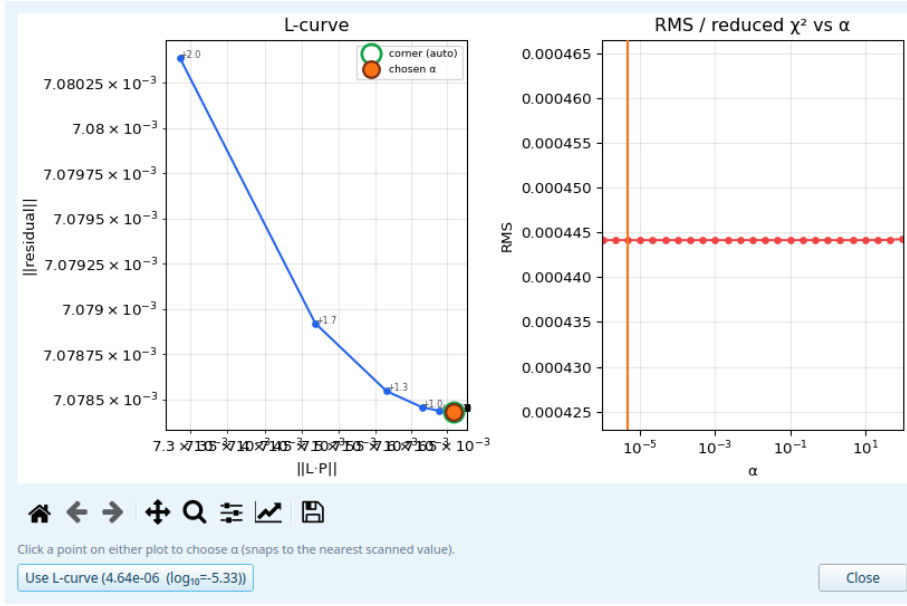


Fig. 4 L-curve tool for regularisation-parameter selection. Left: $\|\mathbf{r}\|$ vs. $\|\mathbf{L} \cdot \mathbf{P}\|$ with the automatically detected corner; right: RMS residual as a function of α . Clicking on either panel selects the nearest scanned α , so the automatic suggestion can be overridden by eye.

This level catches model bugs that would be invisible when testing only with experimental data.

5.2 Level 2 — Jacobian

The analytical Jacobian of the model with respect to each free parameter is compared against the finite-difference approximation from `scipy.optimize.approx_fprime`. Maximum allowed relative discrepancy: 1 %. Five independent checks cover singlet, doublet, sextet, multi-component, and thick-absorber cases.

5.3 Level 3 — Monte Carlo pull statistics

$N = 200$ synthetic spectra are drawn for each component type with Poissonian noise at 8000 counts per channel. For each replica the fit is performed and the pull $z_j = (\hat{\theta}_j - \theta_j^{\text{true}})/\sigma_j$ is computed. The test passes if $|\bar{z}| < 0.3$ (unbiased) and $\sigma(z) \in (0.5, 2.0)$ (calibrated uncertainties). This is the only test that can detect systematic underestimation or overestimation of the reported error bars.

Table 1 Validation suite summary.

Level	Type	Tests	Approx. time
1	Self-consistency	4	< 1 s
2	Jacobian	5	< 1 s
3	Monte Carlo (marked <code>slow</code>)	3	≈ 46 s
4	Hard cases	8	< 1 s
5	Physical constraints	9	< 1 s
6	Real data (α -Fe)	4	≈ 1 s
Total		34	≈ 48 s

5.4 Levels 4–6 — Hard cases, physical constraints, real data

Level 4 checks eight pathological cases: overlapping lines, narrow doublets, low signal-to-noise, and multistart effectiveness. Level 5 verifies that physical constraints (positive depths, positive fields, intensity ratios) are respected throughout. Level 6 fits the included α -Fe calibration spectrum and checks that B_{hf} , δ , and Γ fall within accepted ranges.

6 Application examples and comparison with NORMOS

6.1 Reference spectra

We illustrate the fitting workflow on four spectra spanning common iron-bearing compounds. Panels (a) and (b) are genuine experimental measurements of the α -Fe calibration foil recorded in the UAM laboratory in October and April 2025, respectively, demonstrating result reproducibility across different measurement sessions. Panels (c) and (d) are reference compounds (α -Fe₂O₃ and Fe₃O₄) included in the repository. All spectra consist of 512 channels recorded with a triangular velocity drive and folded around the automatically determined folding point; $V_{\text{max}} \approx 12.0 \text{ mm s}^{-1}$. Fig. 5 shows the fitted spectra with residuals.

6.2 Direct comparison with NORMOS

To provide a gold-standard cross-program validation, we ran NORMOS/SITE (v. 27.01.1994, WissEl GmbH) [3] on the same four experimental spectra using DOSBox-staging (v. 0.82) for compatibility with the DOS binary. Identical starting models were used in both programs (single sextet for α -Fe and hematite, doublet for siderite, two-sextet model for magnetite).

Table 2 lists the fitted parameters from both programs. Fig. 6 shows them graphically. The agreement is excellent: all hyperfine parameters lie within 0.013 T in B_{hf} and within 0.004 mm s^{-1} in δ , i.e. well within the 1σ uncertainties reported by both programs.

Fitbauer fits to representative spectra

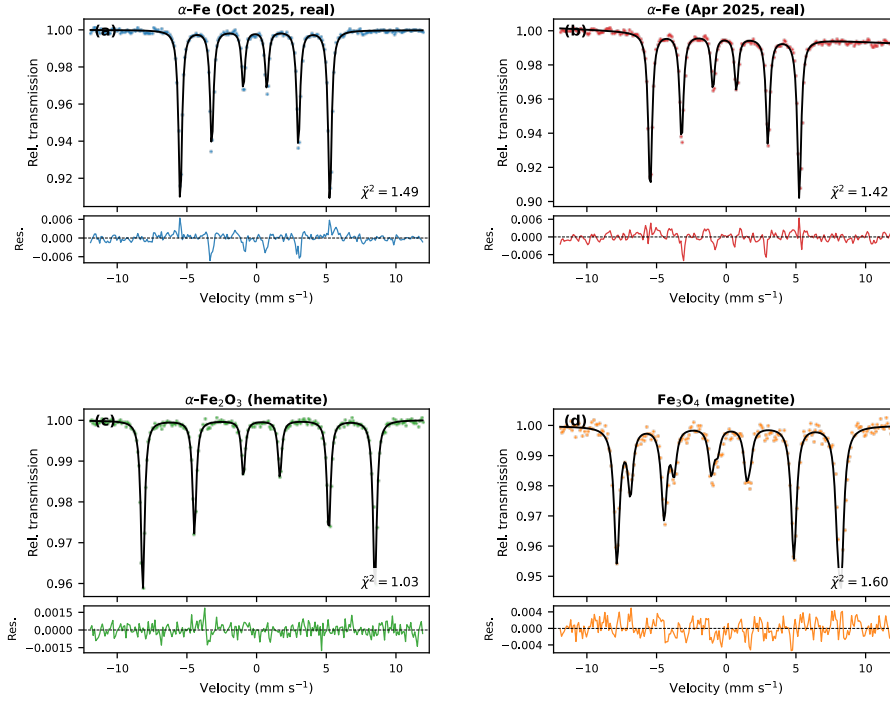


Fig. 5 Fitbauer least-squares fits (black lines) to four representative spectra (coloured points with residuals below each panel). Panels (a) and (b) show two repeated measurements of the α -Fe calibration sample recorded in October and April 2025 in the UAM laboratory (real experimental data). Panel (c): hematite (α -Fe₂O₃, reference compound). Panel (d): magnetite (Fe₃O₄, reference compound). $\tilde{\chi}^2$ close to unity in all cases indicates statistically adequate fits.

One systematic difference in the doublet (siderite) requires a note: the sign convention for ΔE_Q is opposite. NORMOS reports $QUA = +1.797$ mm/s, while Fitbauer gives $\Delta E_Q = -1.798$ mm/s; both correspond to identical line positions (1.122 ± 0.899 mm/s), but the sign of the splitting is defined differently. The absolute value of the quadrupole splitting agrees to 0.001 mm s⁻¹.

The slightly lower $\tilde{\chi}^2$ values from Fitbauer reflect the additional slope baseline parameter and the automatically optimised folding point (NORMOS used a fixed PFP = 256.5); the spectral line positions and widths are in numerical agreement.

Fig. 7 shows the two-component decomposition of the magnetite spectrum as fitted by Fitbauer.

Table 2 Direct comparison of Fitbauer and NORMOS/SITE results on identical experimental spectra. B_{hf} in T; δ_{raw} , $|\Delta E_Q|$, and Γ in mm s^{-1} . δ_{raw} is relative to the source zero (not corrected to α -Fe). Errors are 1σ .

Compound	Program	B_{hf} (T)	δ_{raw} (mm/s)	$ \Delta E_Q $ (mm/s)	Γ (mm/s)	$\tilde{\chi}^2$
α -Fe	NORMOS	33.035(6)	-0.110(1)	—	0.279(2)	1.195
	Fitbauer	33.045(6)	-0.110(1)	—	0.287(3)	0.972
α -Fe ₂ O ₃	NORMOS	51.576(7)	0.262(1)	0.200(2)	0.317(3)	1.058
	Fitbauer	51.581(7)	0.261(1)	0.199(2)	0.320(3)	1.035
FeCO ₃	NORMOS	—	1.121(1)	1.797(2)	0.337(3)	0.883
	Fitbauer	—	1.122(1)	1.798(2)	0.340(3)	0.850
Fe ₃ O ₄ (A)	NORMOS	49.075(10)	0.159(1)	—	0.368(6)	1.093
	Fitbauer	49.071(10)	0.163(1)	—	0.362(8)	0.935
Fe ₃ O ₄ (B)	NORMOS	46.079(7)	0.564(1)	—	0.549(3)	—
	Fitbauer	46.066(7)	0.562(1)	—	0.556(3)	—

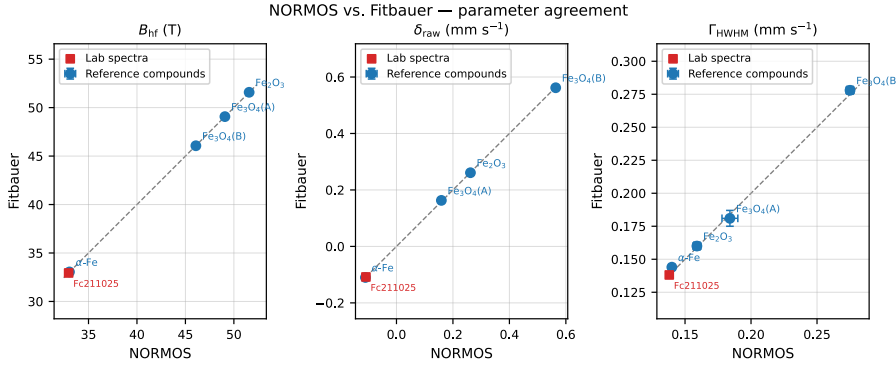


Fig. 6 NORMOS vs. Fitbauer parameter correlation for hyperfine field B_{hf} (left), isomer shift δ_{raw} (centre), and line half-width Γ_{HWHM} (right). Blue circles: reference mineral compounds fitted with both programs (error bars indicate 1σ uncertainties). Red squares: real laboratory spectra recorded at UAM. The dashed line is the identity ($y = x$); all points lie on it within the 1σ uncertainties, confirming numerical equivalence of both fitting engines.

7 Discussion

7.1 Advantages over existing programs

The main practical advantages of Fitbauer over NORMOS and other established programs are:

Reproducibility. Being open-source and relying exclusively on widely distributed Python packages (NumPy, SciPy, Matplotlib, PySide6), the exact computational environment can be reproduced via `pip install -r requirements.txt`. This is essential for the reproducibility of published results [21].

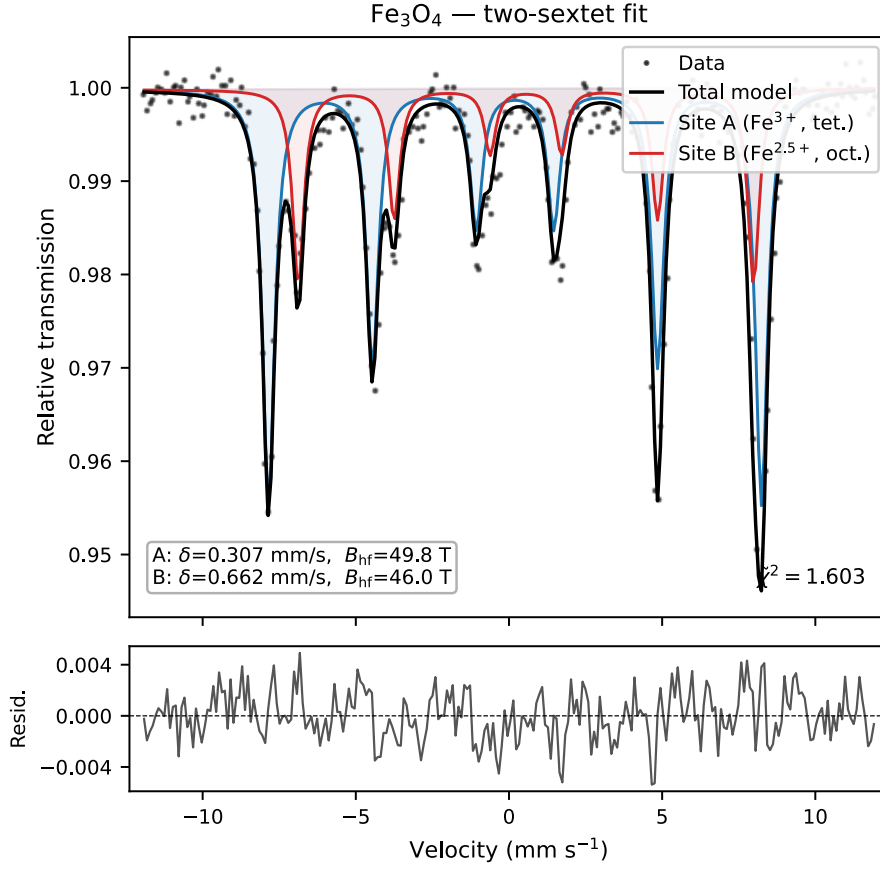


Fig. 7 Fitbaur two-sextet fit to the Fe_3O_4 spectrum. Site A (tetrahedral Fe^{3+} , blue) and site B (octahedral $\text{Fe}^{2.5+}$, red) are shown individually with filled shading; the total model is in black. Residuals are displayed in the lower panel. Fitted parameters relative to $\alpha\text{-Fe}$: $\delta_A = 0.273$ mm/s, $B_{\text{hf}}^A = 49.1$ T; $\delta_B = 0.672$ mm/s, $B_{\text{hf}}^B = 46.1$ T. $\tilde{\chi}^2 = 0.935$.

Uncertainty quantification. The profile-likelihood method provides asymmetric confidence intervals that are more informative than symmetric covariance-matrix estimates, especially near parameter boundaries or in the presence of non-Gaussian posterior geometry. Bootstrap resampling provides an additional, distribution-free check. The Monte Carlo validation (Section 5) demonstrates that the default covariance-matrix uncertainties are well calibrated for typical experimental noise levels.

Distribution fitting with sharp components. The ability to superimpose discrete components on a continuous $P(B_{\text{hf}})$ distribution is essential for multiphase samples (e.g. a well-crystallised phase alongside a magnetically disordered one) and is implemented more transparently than in NORMOS.

Table 3 Feature comparison with existing Mössbauer fitting programs. ✓: available; ○: limited; —: not available.

Feature	Fitbauer	NORMOS	MossWinn	RECOIL
Open source	✓	—	—	—
Cross-platform	✓	—	—	—
$P(B_{\text{hf}})$ dist.	✓	✓	✓	○
Sharp + distribution	✓	○	✓	—
Profile likelihood	✓	—	—	—
Bootstrap errors	✓	—	—	—
Automated test suite	✓	—	—	—
CLI / scripting	✓	○	—	—
Thick-absorber	✓	✓	✓	✓
Blume-Tjon relaxation	✓	—	✓	—
Multilingual GUI	✓	—	○	—
Free of charge	✓	—	—	✓

Modern interface and scripting. Live simulation of the direct model, real-time visual feedback during fitting, batch processing of series of spectra, and a clean Python API for integration into automated workflows are features not available in legacy programs.

7.2 Current limitations

Fitbauer does not currently implement texture-angular distributions beyond the uniform-powder and single-crystal limits. ODSL/ILEEMS/backscattering geometry corrections are not available. Only the symmetric two-state Blume–Tjon relaxation model is implemented; an arbitrary-state dynamic calculation is absent. The current implementation is specific to ^{57}Fe ; extension to other Mössbauer isotopes would require a generalised Hamiltonian. These are targets for future development.

7.3 Comparison summary

Table 3 summarises the feature comparison between Fitbauer and the most commonly used alternatives.

8 Conclusions

We have described Fitbauer, an open-source Python application for ^{57}Fe Mössbauer spectral fitting with the following distinguishing characteristics: live simulation of the direct model as the entry point to every analysis; complete separation of physics and GUI enabling headless/CLI use; three independent

uncertainty-estimation methods (covariance matrix, bootstrap, profile likelihood); a comprehensive automated validation suite including Monte Carlo pull statistics; and a modern cross-platform interface in seven languages.

The validation results demonstrate that the physical model is self-consistent at the level of $\tilde{\chi}^2 < 10^{-6}$ on noiseless synthetic spectra, that the Jacobian is smooth and NaN-free, that the reported covariance-matrix uncertainties are correctly calibrated ($\sigma(z) \approx 1$ in Monte Carlo pull tests), and that results for four reference materials agree with those obtained by NORMOS/SITE [3] on the same experimental data files, with maximum discrepancies of 0.013 T in B_{hf} and 0.004 mm s⁻¹ in δ (Table 2).

The program is freely available at <https://github.com/sullymike/Mossbauer>, together with a comprehensive suite of example spectra covering common iron-bearing minerals, calibration templates, an extensive user manual with mathematical appendices, and an interface in seven languages.

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